

The New Higgs Problem

Sarthak Arora¹

Abstract

This document provides a brief introduction to the off-shell Higgs measurement. We first discuss the importance of the measurement and the process involved before talking about the current status and the future possibilities and challenges.

¹Indian Institute of Science, Bengaluru, India

✉ sarthaka@iisc.ac.in

Introduction

It was 2012 when the Higgs was discovered at CERN and today, almost 13 years later, active research on it is still going on. As it is well known about the role of Higgs in *giving* (almost all) particles their mass via the Higgs Mechanism but there are a number of crucial things that we don't know as of now. One of them is the *Total Width of the Higgs Boson* ($\Gamma = \hbar/\tau$, τ being the lifetime). The total width provides very important information regarding the lifetime of a particle and its interactions with other fields. Using our prior knowledge about Standard Model (SM), we know several channels through which the Higgs interact, but if we take into account some “invisible” contributions then what we have is [1],

$$\Gamma \simeq \Gamma_{SM} + \Gamma_{invisible}$$

Here, $\Gamma_{SM} \simeq 4.1 \text{ MeV}$, as predicted by theory, and is parametrized by the decay products from the various channels of the Higgs. “As Γ represents the sum of the partial widths of the individual decay modes of the Higgs, any modification caused by an additional Beyond SM (BSM) interaction would impact its value. . . this makes the measurement of the width sensitive to the BSM physics” [2]. But we have a problem here. The limited resolution of the ATLAS and CMS detectors doesn't allow us to make precise direct measurements of this width.

1. An Important Observation

But as it is said, “Necessity is the mother of all inventions”, an indirect way to measure the width has been figured out, although it also has its own problems. You see, the Higgs decays into products via several SM interactions. The idea is to study the production and the decay rates of the on-shell and the off-shell Higgs particles and using that we can estimate the

width. Talking about one specific channel, the production of a pair of Vector Boson (WW or ZZ), “this method compares the rates of Higgs boson production measured across two parts of the phase space distinguished by the invariant mass of the vector boson pair, m_{VV} : when m_{VV} is close to the Higgs boson pole mass, m_H (on-shell regime), and when m_{VV} is much greater than that value (off-shell regime)” [2]. The special thing about this is that the signal strength of production of the pair of vector bosons is dependent on the width, in case of the on-shell Higgs decay, whereas it is independent in the off-shell one. In both cases, we have a dependency on the rates of production and decay of the Higgs as well. So,

$$\mu_{\text{on-shell}} = \kappa_{\text{prod,on-shell}}^2 \cdot \kappa_{\text{decay,on-shell}}^2 \cdot \kappa_H^{-1}$$

$$\mu_{\text{off-shell}} = \kappa_{\text{prod,off-shell}}^2 \cdot \kappa_{\text{decay,off-shell}}^2$$

Here, κ s are the ratio of the experimentally measured value to that predicted by the SM theory, i.e. $\kappa_H = \Gamma/\Gamma^{SM}$ and $\kappa_i = g_i/g_i^{SM}$, g_i being the coupling constant with other fields. The decay of the Higgs can happen mainly via gluon-gluon fusion (fermion coupling) or HVV interaction (electroweak).

2. A Way Out

The next idea is to somehow measure the mass of the vector bosons produced so that, using the four-momentum conservation, we can estimate the mass of the Higgs that was involved in the process and calculate the total width. But, you see, the vector bosons themselves decay via two channels, $2l2\nu$ and $4l$ channel ($l = e/\mu$). The former channel is harder to deal with as some momentum information is carried by the neutrinos (ν), which can't be measured. Another very important point before moving ahead is that there exist processes that

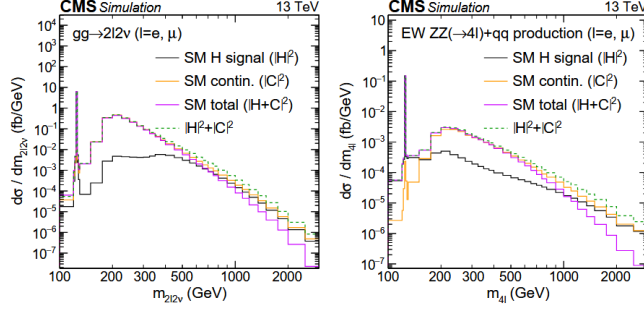


Figure 1. Simulated On and Off-shell regimes [3]

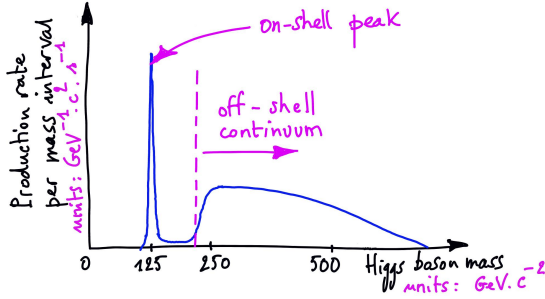


Figure 2. On and Off-shell regimes [4]

have the same initial and final states, like in the Higgs decay. This causes an interference due to the possibility of the initial state going to the final state and involving the Higgs. The interference is destructive, which plays a key role in the SM as it is one of the contributions that unitarizes the scattering of massive gauge bosons, keeping the computation of the cross section for ZZ production in proton-proton (pp) collisions finite” [3]. This can be observed in Figure:1 and Figure:2. The peak refers to the resonant energy at the on-shell mass of the Higgs, while the “hump” in the off-shell regime is a result of the destructive interference. The energy in the hump regime is high enough to produce two real vector bosons. At this stage you should ask a question, “If the energy in the hump regime produces two real vector bosons, then how are produced from the decay of the on-shell Higgs?”. Well, one of the vector boson, in that case, is virtual.

Thus, the processes, we have are:

$$gg \rightarrow H \rightarrow ZZ, \quad qq \rightarrow H \rightarrow ZZjj$$

for the real Higgs and similar for the virtual one. The processes along with the interfering ones are given in Figure:3.

3. Concluding Remarks

The value of the total width came out to be $\Gamma = 4.3^{2.7}_{-1.9} MeV$ as calculated from measurements by ATLAS [5] and $\Gamma = 3.2^{2.4}_{-1.7} MeV$ as calculated from measurements by CMS [6]. Studying the other channel is also particularly relevant as

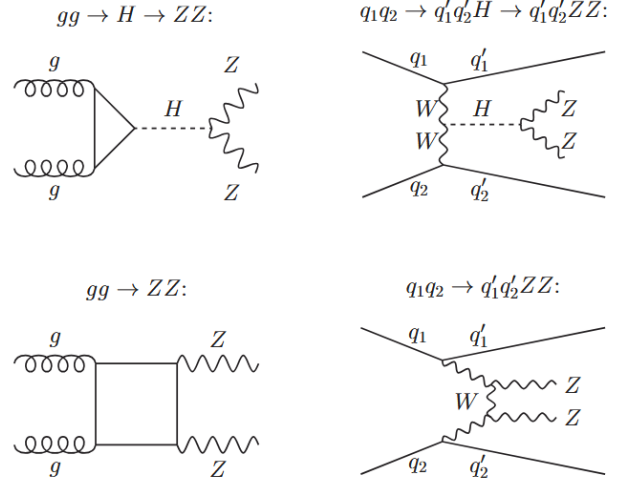


Figure 3. Feynman diagrams for the major processes [3]

BSM scenarios may affect $H \rightarrow WW$ and $H \rightarrow ZZ$ differently. In particular, models that violate custodial symmetry, including scalar extensions of the Standard Model (e.g., multi-Higgs doublet models), can introduce deviations between these channels, making it essential to analyze both in order to fully probe new physics effects [2].

References

- [1] Qing-Hong Cao, Hao-Lin Li, Ling-Xiao Xu, and Jiang-Hao Yu. What can we learn from the total width of the higgs boson?*. *Chinese Physics C*, 47(3):033101, March 2023.
- [2] ATLAS Collaboration. Constraining off-shell higgs boson production and the higgs boson total width using $ww \rightarrow \ell\nu\ell\nu$ final states with the atlas detector, 2025.
- [3] The CMS Collaboration. Measurement of the higgs boson width and evidence of its off-shell contributions to zz production. *Nature Physics*, 18(11):1329–1334, October 2022.
- [4] CMS Collaboration. The life of the higgs boson: <https://cms.cern/news/life-higgs-boson>.
- [5] The ATLAS Collaboration. Measurement of off-shell higgs boson production in the $h^* \rightarrow zz \rightarrow 4\ell$ decay channel using a neural simulation-based inference technique in 13 tev pp collisions with the atlas detector. *Rep. Prog. Phys.*, 88(5):057803, 2025.
- [6] The CMS Collaboration. Measurement of the higgs boson width and evidence of its off-shell contributions to zz production. *Nature Physics*, 18(11):1329–1334, October 2022.